

A Federated Learning Framework for Handling Subtype Confounding and Heterogeneity in Large-Scale Neuroimaging Diagnosis

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Abstract. Computer-aided diagnosis (CAD) systems play a crucial role in analyzing neuroimaging data for neurological and psychiatric disorders. However, small-sample studies suffer from low reproducibility, while large-scale datasets introduce confounding heterogeneity due to multiple disease subtypes being labeled under a single category. To address these challenges, we propose a novel federated learning framework tailored for neuroimaging CAD systems. Our approach includes a dynamic navigation module that routes samples to the most suitable local models based on latent subtype representations, and a meta-integration module that combines predictions from heterogeneous local models into a unified diagnostic output. We evaluated our framework using a comprehensive dataset comprising functional Magnetic Resonance Imaging (fMRI) data from over 1300 Major Depressive Disorder (MDD) patients and 1100 healthy controls (HC) across multiple study cohorts. Experimental results demonstrate significant improvements in diagnostic accuracy and robustness compared to traditional methods. Specifically, our framework achieved an average accuracy of 74.01% across all tested sites, showcasing its effectiveness in handling subtype heterogeneity and enhancing model generalizability. Ablation studies further confirmed the importance of both the dynamic navigation and meta-integration modules in improving performance. By addressing data heterogeneity and subtype confounding, our framework advances reliable and reproducible neuroimaging CAD systems, offering significant potential for personalized medicine and clinical decision-making in neurology and psychiatry.

Keywords: Computer-aided diagnosis, Federated learning, Neuroimaging, Functional magnetic resonance imaging, Major depressive disorder

1 Introduction

Computer-aided diagnosis (CAD) has become an indispensable tool for functional magnetic resonance imaging (fMRI) analysis, enabling the identification of complex brain activity patterns associated with neurological and psychiatric disorders [1–3]. As neuroimaging moves toward more data-driven and personalized paradigms, CAD systems increasingly support early detection, differential diagnosis, and treatment monitoring [3, 4].

A major limitation in current neuroimaging research is the reliance on small-sample studies, which often suffer from low statistical power and poor reproducibility [5]. It has been widely recognized that robust and reproducible brain-wide association studies require datasets encompassing thousands of individuals [6]. While increasing the sample size enhances statistical reliability, it also introduces new challenges. Notably, aggregating data across multiple institutions or populations often results in increased heterogeneity, for instance, a single diagnostic label may encompass multiple subtypes of a disorder [7]. This confounding variability can mislead conventional CAD models, leading to reduced generalizability and compromised diagnostic performance.

To mitigate heterogeneity in fMRI-based CAD, prior works typically treat site information and individual differences as confounders and attempt to suppress them. Representative strategies include adversarial separation between task-relevant features and confounders [8], attention mechanisms guided by non-imaging representations to reduce individual variability [9], and statistical harmonization methods such as Subsampling Maximum-mean-distance and CovBat to correct site-specific effects [10]. Domain adaptation methods further map data into a target domain to alleviate distribution mismatch [11, 12]. Despite these advances, most approaches do not explicitly address subtype heterogeneity, and often rely on manually designed confounder representations or require a pre-defined target domain. This limitation is particularly evident in disorders such as MDD, where complex subtype structures can introduce significant ambiguity for CAD systems [4].

Federated learning (FL) provides a promising privacy preserving alternative for multi-site neuroimaging analysis by enabling collaborative training without centralizing sensitive data [13–15]. In FL, each site performs local training and shares only model updates with a server [16, 17], which has shown strong potential in medical CAD applications under privacy regulations [18]. To handle inter-site heterogeneity, personalized federated learning (PFL) methods learn client-specific models and deploy local classifiers [19–22]. However, most existing PFL approaches personalize at the client level and implicitly assume intra-site homogeneity, which limits adaptability when subtype variability exists within a site and becomes especially restrictive for sample-level inference.

In this work, we propose a federated learning framework tailored for neuroimaging CAD that addresses two core challenges under subtype heterogeneity: (i) navigation, determining which local model is most appropriate for each sample when labels are shared but subtypes differ; and (ii) integration, combining heterogeneous local models into a unified and robust diagnostic system. Our approach introduces a dynamic routing mechanism to assign samples to suitable local models based on latent subtype representations, and a meta-integration strategy to fuse predictions across models into a coherent output. Furthermore, we develop a two-stage framework based on Adaptive Attention Aggregation : the first stage learns site features and performs heterogeneous classification to capture cross-site differences, while the second stage adopts attention-based

adaptive aggregation across site models to achieve personalized prediction at the individual sample level. Our main contributions are summarized as follows:

- **Dynamic Navigation Mechanism:** We propose a dynamic routing module that assigns samples to the most suitable local models for training based on latent subtype representations.
- **Meta-Integration Strategy:** We introduce a meta-integration approach that effectively combines predictions from heterogeneous local models into a unified diagnostic output with enhanced consistency.
- **Two-Stage Framework:** We develop a two-stage framework for brain image classification: site feature learning with heterogeneous classification, followed by attention-based cross-site adaptive aggregation for sample-level personalized prediction.

2 Methodology

2.1 Overview

We propose a two-stage framework for brain image classification that addresses data heterogeneity in multi-center medical data and explores the impact of disease subtypes on diagnostic accuracy. Our method enables personalized, sample-level predictions through heterogeneous modeling and adaptive fusion, as illustrated in Figure 1.

By integrating subtype-aware modeling and attention-driven aggregation, our method effectively mitigates the impact of data heterogeneity in multi-center settings. The results demonstrate that modeling subtypes improves classification accuracy, and the proposed framework significantly enhances model performance through personalized, context-aware inference.

2.2 Stage I: Site feature learning and heterogeneous classification

To effectively address the inherent data heterogeneity issue in multi-site medical data, enhance the local classification performance of each site, and gain a deeper understanding of the data differences between sites, we have designed a site feature learning and heterogeneous classification module. This module receives labeled datasets $\{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_N\}$ from N different sites as input. The core architecture consists of a set of homogeneous autoencoders $\{E_1, E_2, \dots, E_N\}$ and a set of heterogeneous classifiers $\{C_1, C_2, \dots, C_N\}$. Through the homogeneous autoencoder, we obtain the template for each site, including the MDD template and the HC template. Finally, the heterogeneous classifier C_s , homogeneous autoencoder E_s , and template T_s for site s are uploaded to the server. The server aggregates the homogeneous autoencoders from each site with weights to obtain the global homogeneous autoencoder E_g , while storing the heterogeneous classifiers and template for each site. The workflow of Stage I is shown in the upper part of Figure 1.

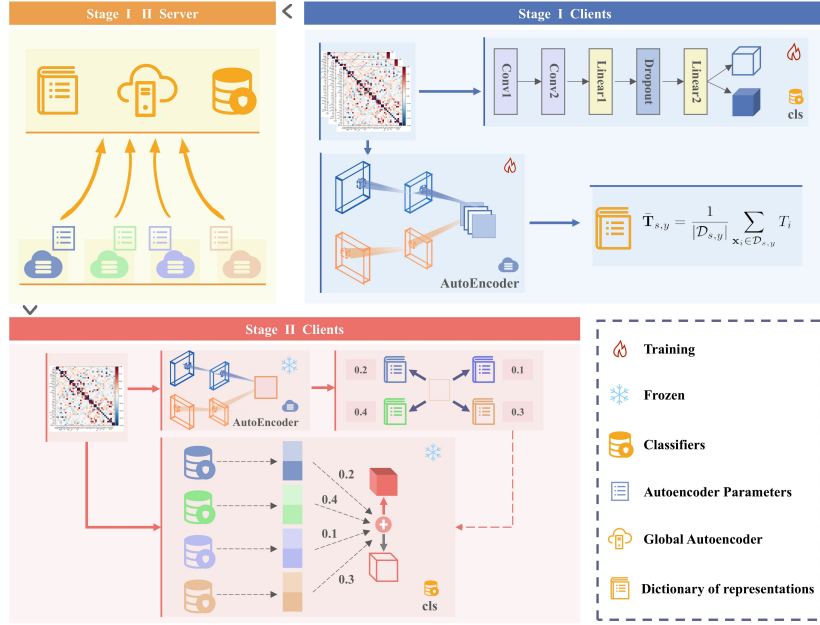


Fig. 1. Our method framework consists of three main parts: (stage I clients) each client trains local autoencoders and classifiers, calculates representation dictionaries, and uploads them to the server; (Stage I II server) The server-side stores client parameters and dictionaries, and calculates global autoencoder parameters; (Stage II clients) Each client updates model parameters, calculates site similarity and prediction values for each test data, and finally weights them to obtain the final prediction value.

By utilizing the symmetry of the FC matrix to extract the upper triangular elements and construct feature vectors as input, the encoder part of the autoencoder maps the feature vectors to the latent feature space, obtains the low dimensional representation of the corresponding label, and averages it, further obtain the template for the label, and the mapping process can be formalized as follows:

$$\bar{T}_{s,y} = \frac{1}{|\mathcal{D}_{s,y}|} \sum_{\mathbf{x}_i \in \mathcal{D}_{s,y}} T_i, T_i = \mathcal{E}_\theta(\mathbf{x}_i), y \in \{0, 1\}, \quad (1)$$

where $\mathcal{D}_{s,y}$ denotes all sample sets with label y at site s , $\mathcal{E}_\theta$ denotes the autoencoder with parameter θ , $\mathbf{x}_i$ denotes the upper triangular flattened vector of a single sample, $\bar{T}_{s,y}$ denotes the latest template for the sample with label y at site s .

The server aggregates the autoencoders of each site to obtain a global autoencoder. Aggregation process is as follows:

$$\theta^{\text{global}} = \sum_{s=1}^N w_s \theta^{(s)}, w_s = \frac{|\mathcal{D}_s|}{\sum_{k=1}^N |\mathcal{D}_k|}, \quad (2)$$

where w_s denotes Weight of site s , $\theta^{(s)}$ denotes local parameters of site s autoencoder, θ^{global} denotes parameters of autoencoder after global aggregation.

In this module, each site trains local heterogeneous classifiers and homogeneous autoencoders with local data to obtain MDD template and HC template, and uploads heterogeneous classifier C_i , homogeneous autoencoder E_i , MDD template $\bar{\mathbf{T}}_{s,0}$, and HC template $\bar{\mathbf{T}}_{s,1}$ to the server. The server weight aggregates the homogeneous autoencoders of each site to obtain global autoencoder E_g , and distributes the global autoencoder E_g , heterogeneous classifier C_i , MDD template $\bar{\mathbf{T}}_{s,0}$ and HC template $\bar{\mathbf{T}}_{s,1}$ to each site, effectively addressing the heterogeneity challenges in multi-center medical data analysis, improving the classification accuracy of the model, and enhancing the understanding of data distribution differences.

2.3 Stage II: Cross-site model adaptive aggregation based on attention mechanism

Firstly, the autoencoder E_g trained in Stage I is used to process the upper triangular flattening vector of a single sample, obtaining the low dimensional representation T of the sample. Then, the FC is input to heterogeneous classifiers at each site, and the predicted logits $\mathbf{l}$ on each site classifier are obtained separately, calculate the attention score of each site, weight the logits of each site based on the attention score, and obtain the final logits value to achieve personalized prediction of a single sample. The workflow of Stage II is shown in the upper part of Figure 1.

Calculate the cosine similarity between the low dimensional representation T of the sample and the two templates ($\bar{\mathbf{T}}_{s,0}$ and $\bar{\mathbf{T}}_{s,1}$) at site s . Calculation process can be formalized as follows:

$$\cos(T, \bar{\mathbf{T}}_{s,y}) = \frac{T^\top \bar{\mathbf{T}}_{s,y}}{\|T\| \|\bar{\mathbf{T}}_{s,y}\|} = \frac{\sum_{i=1}^d T_i \bar{\mathbf{T}}_{s,y,i}}{\sqrt{\sum_{i=1}^d T_i^2} \sqrt{\sum_{i=1}^d \bar{\mathbf{T}}_{s,y,i}^2}}, \quad (3)$$

where d denotes the number of elements in the upper triangle, $\sum_{i=1}^d T_i \bar{\mathbf{T}}_{s,y,i}$ denotes the dot product of low dimensional representation T and template $\bar{\mathbf{T}}_{s,y}$, used to measure the degree of alignment between the two in the direction, $\sqrt{\sum_{i=1}^d T_i^2}$ denotes the euclidean norm of low dimensional representation T , $\sqrt{\sum_{i=1}^d \bar{\mathbf{T}}_{s,y,i}^2}$ denotes the euclidean norm of template $\bar{\mathbf{T}}_{s,y}$.

To obtain the weights of predicted logits for each site, the original attention scores for each site are first calculated based on similarity, and then normalized. Similarity calculation aims to quantify the similarity between the data features of the target sample and the overall distribution features of each locus, where higher similarity indicates better matching between the sample and the corresponding locus. Specifically, each site provides two templates $\bar{\mathbf{T}}_{s,0}$ and $\bar{\mathbf{T}}_{s,1}$, and the similarity between each sample and the two templates of different sites is calculated to obtain the original attention score of that site. The final prediction weight is obtained by normalizing the original attention scores of all sites.

Therefore, the closer the templates of a site are to the target sample, the greater its weight and contribution to the final decision. The calculation process is as follows:

$$w_{l,s} = \frac{\alpha_s}{\sum_{j=1}^N \alpha_j}, \alpha_s = \cos(T, \bar{\mathbf{T}}_{s,0}) + \cos(T, \bar{\mathbf{T}}_{s,1}), \quad (4)$$

Multiply the attention scores w_s of each site by their corresponding predicted logits $\mathbf{l}_s$ and add them together to obtain the final predicted logits $\hat{\mathbf{l}}$, $\hat{\mathbf{l}} = \sum_{s=1}^N w_{l,s} \mathbf{l}_s$.

Based on the characteristics of the target sample, dynamically fuse model information from all sites, prioritize the selection and fusion of model decisions from sites with the most similar data distribution to the target sample, effectively alleviate the distribution offset caused by data heterogeneity, and significantly improve the accuracy and robustness of model. Compare the difference between predicted labels and real labels to calculate the accuracy of the test. This process is independently executed for each input sample, achieving highly personalized prediction.

In this stage, the matching degree between each test sample of the site and the data characteristics of each site is calculated, and attention scores are generated. The final prediction logits is generated by weighted aggregation based on the attention scores, achieving personalized prediction at the single sample level and effectively improving the prediction accuracy.

3 Results

3.1 Experiment Setup

We conduct experiments on the REST-meta-MDD dataset [23], a large-scale multi-center resting-state fMRI collection from 17 Chinese hospitals, comprising 1,300 patients with MDD and 1,128 HC across 25 cohorts. The multi-site nature of the dataset captures substantial real-world heterogeneity, making it suitable for evaluating federated learning under diverse clinical conditions.

To investigate neurobiological differences across MDD subtypes, patients are stratified along two clinically relevant dimensions: episodicity (first-episode vs. recurrent) and medication status (medicated vs. drug-naïve). Based on this stratification, four subgroups with distinct clinical profiles are defined: Site 1 - recurrent unmedicated MDD (n=76), Site 2 - recurrent medicated MDD (n=121), Site 3 - first-episode drug-naïve (FEDN) MDD (n=318), and Site 4 - first-episode medicated MDD (n=160). Each patient group is matched with an equal number of healthy controls randomly sampled from the HC cohort, resulting in site sizes of 152, 242, 636, and 320 samples, respectively.

For model construction, each site has a heterogeneous classifier, which is closer to the real usage scenario where the computing power, data distribution, and actual business requirements of each site are different. In addition, we also conducted comparative experiments using homogeneous classifiers at four sites.

3.2 Evaluation Results

To evaluate the effectiveness of our method, we compare it with five established federated learning methods: FedAvg [16], FedProto [24], pFedAFM [19], PerAda [25], and Fedgh [26]. Although these baselines were originally evaluated on benchmarks like CIFAR-10 and MNIST, we re-implement and re-train them on the REST-meta-MDD dataset using the same data partitioning and preprocessing pipeline to ensure a fair comparison. And in the field of deep learning, we also investigated five methods: BrainGNN [27], STAGIN [28], STCAL [29], MSSTAN [30], DHGCN [31], all of which were conducted on the REST-meta-MDD. Table 1 reports the Accuracy, Area Under Curve, Sensitive, Specificity, and F1 score of each method.

Table 1. Accuracy of Methods in the Field of Federated Learning

Field	Method	ACC	AUC	SEN	SPE	F1
Federated Learning	FedAvg	68.96 $\pm$ 6.22	76.54 $\pm$ 3.03	67.06 $\pm$ 7.51	70.91 $\pm$ 5.46	67.83 $\pm$ 5.76
	FedProto	61.28 $\pm$ 5.07	66.87 $\pm$ 6.11	62.68 $\pm$ 9.65	59.78 $\pm$ 6.39	60.09 $\pm$ 5.95
	pFedAFM	61.69 $\pm$ 7.64	67.97 $\pm$ 7.13	61.81 $\pm$ 10.84	63.03 $\pm$ 1.78	61.25 $\pm$ 7.11
	PerAda	60.26 $\pm$ 2.74	63.95 $\pm$ 3.37	57.68 $\pm$ 4.42	62.79 $\pm$ 2.12	59.83 $\pm$ 2.86
	Fedgh	55.46 $\pm$ 8.68	58.31 $\pm$ 8.97	54.20 $\pm$ 18.79	57.34 $\pm$ 29.92	48.65 $\pm$ 8.46
Deep Learning	BrainGNN	67.56 $\pm$ 4.57	66.55 $\pm$ 4.86	67.61 $\pm$ 4.93	68.10 $\pm$ 6.41	62.26 $\pm$ 9.79
	STAGIN	66.56 $\pm$ 0.34	70.69 $\pm$ 0.01	68.03 $\pm$ 2.21	64.57 $\pm$ 3.05	68.03 $\pm$ 2.21
	STCAL	66.98 $\pm$ 0.70	73.31 $\pm$ 0.01	70.24 $\pm$ 1.22	63.32 $\pm$ 1.53	70.24 $\pm$ 1.22
	MSSTAN	68.60 $\pm$ 0.90	67.06 $\pm$ 1.39	74.67 $\pm$ 3.26	59.45 $\pm$ 4.84	71.63 $\pm$ 1.23
	DHGCN	68.60 $\pm$ 0.90	71.80 $\pm$ 0.01	71.00 $\pm$ 1.30	65.80 $\pm$ 1.10	71.20 $\pm$ 0.90
Ours	homo	73.93 $\pm$ 2.20	74.61 $\pm$ 3.48	79.36 $\pm$ 3.94	66.28 $\pm$ 3.27	73.85 $\pm$ 1.95
	hetero	74.01 $\pm$ 1.68	74.77 $\pm$ 2.69	78.75 $\pm$ 3.12	66.98 $\pm$ 3.36	73.71 $\pm$ 1.18

To evaluate the contribution of key components in our method, we conduct an ablation study by comparing four variants in MDD diagnostic classification. The complete model incorporates both subtype-based grouping and the MoE mechanism for dynamic, personalized classifier fusion. We then consider a variant that retains subtype classification but removes the MoE module, relying instead on hard selection of the most similar site. A third variant disables subtype grouping (treating all patients uniformly) but keeps the MoE for adaptive fusion. Finally, we evaluate a baseline that removes both subtype classification and MoE, resulting in a static aggregation without personalization. This setup allows us to isolate the effects of clinical subtyping and adaptive attention aggregation on model performance.

3.3 Analysis

Comparative Experiment. As shown in Table 1, our method achieves the highest average classification accuracy among all evaluated methods. Compared

Table 2. ACC of the Ablation Study in the MDD Diagnostic Classification

Subtype	MoE	ACC	AUC	SEN	SPE	F1
✓	✓	74.01 ± 1.68	74.77 ± 2.69	78.75 ± 3.12	66.98 ± 3.36	73.71 ± 1.18
	×	70.01 ± 1.42	70.85 ± 5.82	67.17 ± 9.72	73.47 ± 1.59	68.29 ± 5.44
×	✓	68.04 ± 1.06	65.39 ± 0.56	64.92 ± 10.09	68.09 ± 7.36	63.93 ± 3.69
	×	66.10 ± 2.28	62.87 ± 2.68	59.48 ± 5.12	67.16 ± 8.9	59.1 ± 3.13

to the best-performing federated learning baseline, our method improves accuracy by 5.05%; against the top deep learning baseline, the improvement reaches 5.41%. These consistent gains demonstrate the effectiveness of our framework in leveraging subtype-specific patterns for personalized prediction. The results indicate that incorporating clinical subtyping and adaptive aggregation enables our method to better handle data and model heterogeneity, leading to significantly enhanced performance in multi-site medical image classification.

Ablation Experiment. As shown in Table 2, the complete our model incorporating both subtype classification and the MoE based dynamic fusion achieves the highest performance among all ablation variants. It outperforms the model with subtype classification but without MoE, as well as the model that uses MoE without subtype stratification. This consistent improvement demonstrates that both components contribute independently and positively to classification accuracy. The results suggest that clinical subtype modeling provides a meaningful patient stratification, while the MoE module enables more adaptive and personalized prediction by dynamically weighting local models. Together, they enable a more effective federated framework for heterogeneous medical data.

4 Conclusion

Our study demonstrates that our method, which integrates subtype classification with a MoE module for dynamic, personalized classifier fusion, achieves superior performance in multi-site MDD diagnostic classification. By leveraging clinical subtypes and adaptive aggregation, the complete model significantly outperforms baseline methods that either omit subtype stratification or lack the MoE mechanism. Specifically, our approach improves accuracy by 5.05% over the best federated learning baseline and by 5.41% over the best deep learning baseline. These findings highlight the importance of incorporating both patient-specific characteristics and adaptive modeling strategies to enhance predictive performance in heterogeneous datasets. Future work will focus on extending this framework to other psychiatric disorders and exploring additional methods for improving robustness and generalizability across diverse clinical settings.

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